

The current flowing through a circuit is
 $i(t) = 25 \sin(\omega t - 30^\circ)$ when a voltage of
 $V(t) = 100 \sin \omega t$ is applied. Find active &
 reactive power.

Soln

$$\text{Active Power} = \frac{V_m I_m}{2} \cos \phi$$

~~$$= \frac{25 \sin(\omega t - 30^\circ) \times 100 \sin \omega t}{2}$$~~

$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos 30$$

$$= \frac{100}{\sqrt{2}} \times \frac{25}{\sqrt{2}} \cos 30$$

$$= 1250 \cos 30 = \underline{\underline{1082.53 \text{ W}}}$$

$$\text{Reactive Power} = \frac{V_m I_m}{2} \sin \phi$$

$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \sin \phi$$

$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \sin \phi$$

$$= \frac{100}{\sqrt{2}} \frac{25}{\sqrt{2}} \sin 30 = \underline{\underline{625 \text{ VAR}}}$$

2. A non inductive resistor connected in series with a
 choke coil is fed from a 250V, 50Hz supply. The
 voltage across resistor is 125V & voltage across ^{choke} coil is 200V.
 Current flowing through circuit is 5A. Calculate

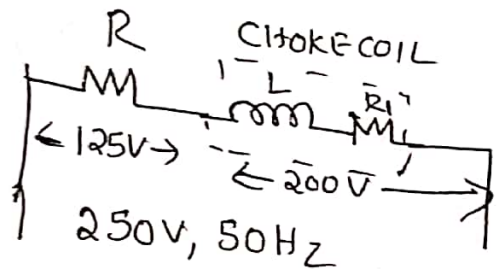
(i) Reactance of coil.

(ii) The inductance of the coil

~~(iii) Voltage across the choke coil.~~ (iii) Voltage across choke coil.

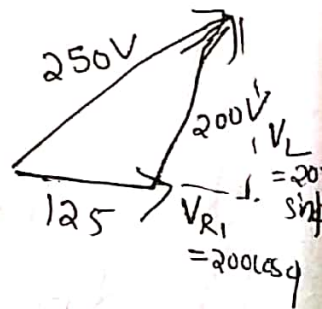
(iv) Power absorbed by choke coil.

$$R = \frac{V_R}{I} = \frac{125}{5} = 25 \Omega$$



$$Z_{\text{Total}} = \frac{V}{I} = \frac{250}{5} = 50 \Omega$$

$$Z_{\text{choke coil}} = \frac{V_{\text{choke coil}}}{I} = \frac{200}{5} = 40 \Omega$$



$$R_{cc}^2 + X_L^2 = Z_{cc}^2$$

$$R_{cc}^2 + X_L^2 = (40)^2$$

$$R_{cc}^2 + X_L^2 = 1600 \quad \text{--- (1)}$$

$$Z_{\text{Total}}^2 = (R + R_{cc})^2 + X_L^2$$

$$50^2 = (25 + R_{cc})^2 + X_L^2 \quad \text{--- (2)}$$

$$2500 = 625 + R_{cc}^2 + 50R_{cc} + X_L^2$$

$$R_{cc}^2 + X_L^2 = 2500 - 625 - 50R_{cc}$$

$$R_{cc}^2 + X_L^2 = 1875 - 50R_{cc} \quad \text{--- (2)}$$

Sub (1) in
(2) $\Rightarrow$

$$1600 = 1875 - 50R_{cc}$$

$$50R_{cc} = 1875 - 1600$$

$$R_{cc} = \frac{275}{50} = 5.5$$

(i) Reactance of coil / $\therefore X_L = \sqrt{1600 - (5.5)^2}$
 $= \underline{\underline{39.62 \Omega}}$

(ii) Inductance of coil, $L = \frac{X_L}{2\pi f}$
 $= \frac{39.62}{2 \times 3.14 \times 50}$
 $= 0.126 \text{ H}$

(iii) Power absorbed by choke coil $= I^2 R_{cc}$
 $= 5^2 \times 5.5$
 $= \underline{\underline{137.5 \text{ W}}}$

3. A capacitor having a capacitance of $20\mu\text{f}$ is connected in series with a non inductive resistance of 200Ω a 220V , 50Hz supply. Calculate.

1) Impedance 2) Current 3) P.F. 4) Power drawn from supply.

Soln

$$V = 220\text{V}, R = 50, R = 200, C = 20 \times 10^{-6}.$$

Soln

$$Z = \sqrt{R^2 + X_C^2} = \sqrt{200^2 + 159.15^2}$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2 \times 3.14 \times 50 \times 20 \times 10^{-6}} = 159.15\Omega$$

$$(i) Z = \sqrt{200^2 + 159.15^2} = 255.6\Omega$$

$$(ii) I = \frac{V}{Z} = \frac{220}{255.6} = 0.86\text{A}$$

$$(iii) \cos\phi = \frac{R}{Z} = \frac{200}{255.6} = 0.7824$$

(iv) Power drawn from the supply

$$P = VI \cos\phi$$

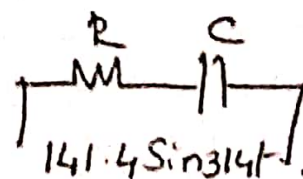
$$= 220 \times 0.86 \times 0.7824$$

$$= \underline{\underline{147.95\text{W}}}$$

4. Find the circuit constants of a 2 element series circuit which consumes 700W with 0.707 lag. The applied voltage is $V = 141.4 \sin 314t$.

$$P = 700\text{W}, \cos\phi = 0.707$$

$$V = 141.4 \sin 314t$$



$$V_{rms} = 141.4$$

$$V_{rms} = \frac{141.4}{\sqrt{2}} = 99.98$$

$$P = V_{rms} I_{rms} \cos \phi$$

$$700 = 99.98 \times I_{rms} \times 0.707$$

$$I_{rms} = \frac{700}{99.98 \times 0.707} = 9.9 \text{ A.}$$

$$R = ?$$

$$C = ?$$

$$Z = \frac{V}{I} = \frac{99.98}{9.9} = 10.098$$

$$\cos \phi = R/Z$$

$$Z = R \cos \phi$$

$$R = Z \cos \phi = 10.098 \times 0.707 = 7.139$$

$$X_C = \sqrt{Z^2 - R^2} = \sqrt{(10.098)^2 - (7.139)^2}$$

$$= 7.135 \text{ } \Omega$$

$$X_C = \frac{1}{2\pi f C}$$

$$C = \frac{1}{2 \times 314 \times 50 \times 7.135}$$

$$= 4.463 \times 10^{-4} \text{ F}$$

The voltage across and current through a circuit are given by

$$v = 250 \sin(314t - 10^\circ) \quad ; \quad i = 10 \sin(314t + 5^\circ)$$

Calculate impedance, resistance & reactance of circuit

Soln

$$V = 250 \sin(314t - 10^\circ)$$

$$i = 10 \sin(314t + 50^\circ)$$

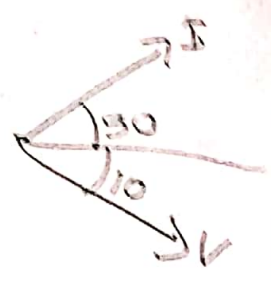
$$\phi = 60^\circ$$

$$V_m = 250$$

$$I_m = 10$$

$$\omega = 314$$

$$2\pi f = 314$$



$$Z = \frac{V}{I}$$

$$V = \frac{V_m}{\sqrt{2}} \angle -10^\circ$$

$$I = \frac{I_m}{\sqrt{2}} \angle 50^\circ$$

$$= \frac{\frac{250}{\sqrt{2}} \angle -10^\circ}{\frac{10}{\sqrt{2}} \angle 50^\circ}$$

$$= 25 \angle -10 - 50$$

$$= 25 \angle -60$$

~~It indicates~~ ^{Here} I leads voltage (capacitive)

$$PF = \cos 60 = 0.5$$

$$\cos \phi = R/Z$$

$$0.5 = \frac{R}{25}$$

$$R = 0.5 \times 25 = \underline{12.5}$$

$$Z = \sqrt{R^2 + X_c^2}$$

$$25^2 = \sqrt{12.5^2 + X_c^2}$$

$$X_c^2 = \sqrt{25^2 - 12.5^2} = 21.5$$

4. A series RLC circuit with a resistance of 50Ω , a capacitance of $25\mu F$ and an inductance of $0.15H$ is connected across $230V$, $50Hz$ supply. Determine the impedance of the circuit?

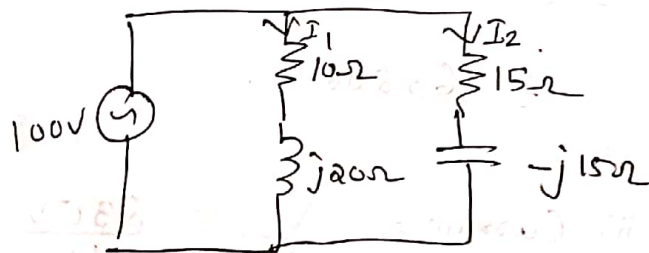
Soln: $R = 50\Omega$, $L = 0.15H$, $C = 25\mu F$.

$$X_L = 2\pi fL \\ = 2\pi \times 50 \times 0.15 = 47.1\Omega$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 25 \times 10^{-6}} = 127.38\Omega$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2} = \sqrt{50^2 + (127.38 - 47.1)^2} \\ = 94.57\Omega$$

5. Find Z_{eq} for parallel circuit shown below. Also find total current & branch current.



$$Z_{eq} = \frac{(10 + j20) \times (15 - j15)}{(10 + j20) + (15 - j15)}$$

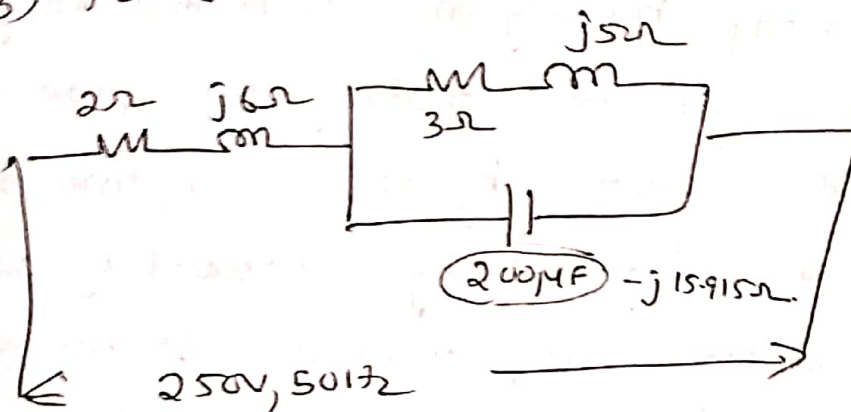
$$= 18.6 \angle 7.13^\circ$$

$$I = \frac{V}{Z} = \frac{100}{18.6 \angle 7.13^\circ} = 5.37 \angle -7.13^\circ$$

$$I_1 = \frac{V}{Z_1} = \frac{100}{10+j20} = 4.472 \angle -63.43^\circ$$

$$I_2 = \frac{V}{Z_2} = \frac{100 \angle 0}{21.21 \angle -45} = 4.72 \angle 45^\circ$$

6. A voltage of 250V at 50Hz frequency is applied to the circuit. Find (i) current drawn from source (2) powerfactor of circuit (3) Power consumed.



$$Z_1 = (3+j5)\Omega$$

$$Z_2 = -j \frac{1}{X_C} = \frac{-j}{\frac{2\pi \times 50 \times 200 \times 10^{-6}}{}} = -j15.915\Omega$$

$$Z_3 = 2+j6$$

$$Z = (2+j6) + \frac{(3+j5) \times -j15.915}{(3+j5) + -j15.915}$$

$$= 7.9115 + j11.6432 = 14.07 \angle 55.8^\circ$$

$$I = \frac{V}{Z} = \frac{250 \angle 0}{14.067 \angle 55.8} = 9.98 - j14.68j$$

$$= 17.76 \angle -55.8^\circ$$

$$\text{Power factor} = \cos \phi = \cos 55.8^\circ = 0.562$$

$$\text{Total power} = VI \cos \phi = 250 \times 17.76 \times 0.562$$

$$= 2495.72 \text{ W}$$

$$= I^2 R = (17.76)^2 \times 7.915 = \underline{\underline{2495.72 \text{ W}}}$$

Three phase System

The single phase generator producing single phase supply has only one armature winding. A two phase generator producing two phase voltages has two windings displaced by 90° & a three phase generator has three windings displaced by 120° . If no. of armature windings increased, it becomes a polyphase system. In general, a polyphase system has many phases, each phase having a single alternating voltage of equal magnitude & frequency but displaced from one another by equal electrical angles.

Advantages of three phase system

1. For a given size of frame, a three phase generator or motor has greater output than that of single phase generator.
2. Three phase generators work in parallel without any difficulty.

3. Three phase transmission line requires less amount of conductor material for transmitting same amount of power over a single phase line.
4. Three phase motors possess uniform torque whereas single phase motors possess a pulsating torque.
5. Three phase induction motors are self starting whereas single phase motors are not self starting.
6. Three phase induction motor is cheaper and efficient than single phase induction motor.

Generation of three phase ac voltage

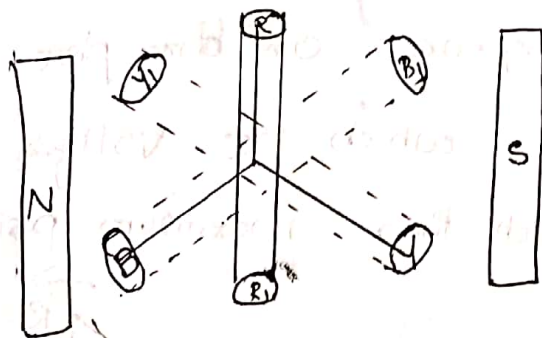
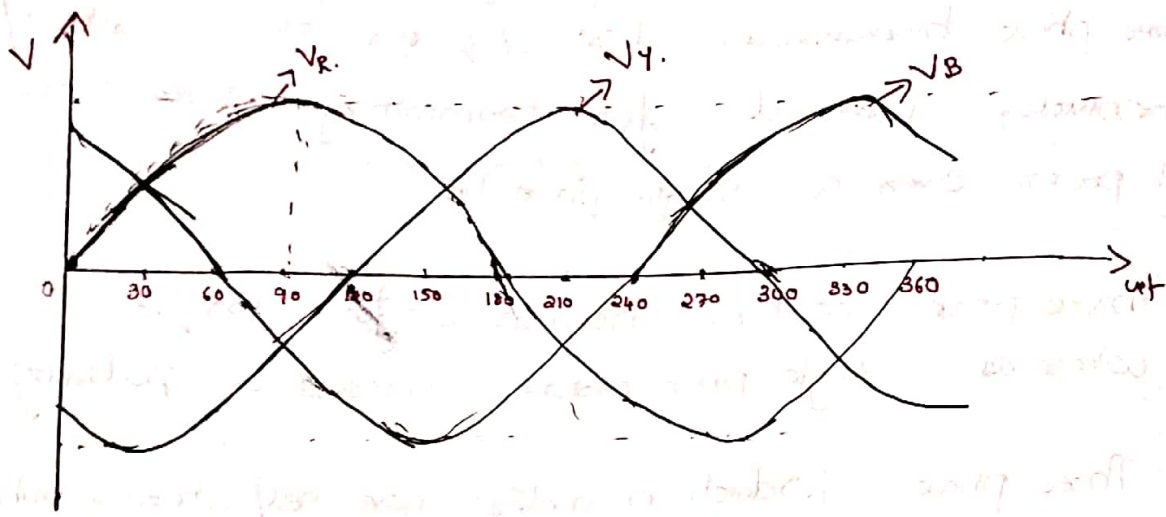


Figure shows an elementary three phase alternator. The three identical coils R, Y, B are placed at 120° from each other and rotated in a uniform magnetic field, sinusoidal voltage is induced in each coil will be of same magnitude & frequency.



The equations of the three induced voltages are .

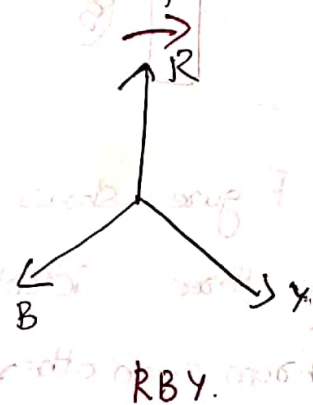
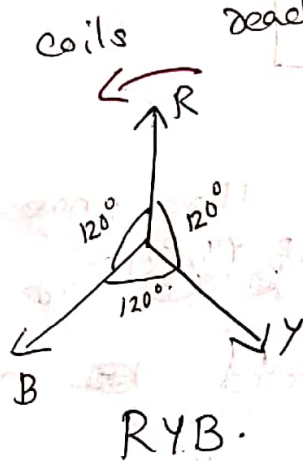
$$V_R = V_m \sin \omega t .$$

$$V_Y = V_m \sin (\omega t - 120^\circ)$$

$$V_B = V_m \sin (\omega t - 240^\circ) = V_m \sin (\omega t + 120^\circ)$$

Phase Sequence

The phase sequence of three phase system gives the order in which the voltages in three phase or coils reach their maximum positive value.



Double Subscript Notation

In this system, two letters are used as subscripts of the symbol for voltage/current

The order of writing the subscripts indicates the direction in which voltage acts on the current flow.

The representation V_{RY} means that the voltage from point R to Y i.e. point R is at a higher potential as compared to the point Y. Similarly the representation I_{RY} means that the current flows from point R to point Y.

$$V_{RY} = -V_{YR}, \quad I_{RY} = -I_{YR}.$$

Symmetrical & balanced 3 ϕ system

A 3 ϕ system is said to be symmetrical when the various voltages are equal in magnitude and are displaced from one another by equal angles.

The system is balanced when the various voltages are equal in magnitude, the various currents are equal in magnitude & the phase angle are the same for each phase.

Methods of connection of 3 ϕ system

There are two methods for

interconnection -

- (1) Star γ connection.
- (2) Delta Δ connection.

I) Star Y connection

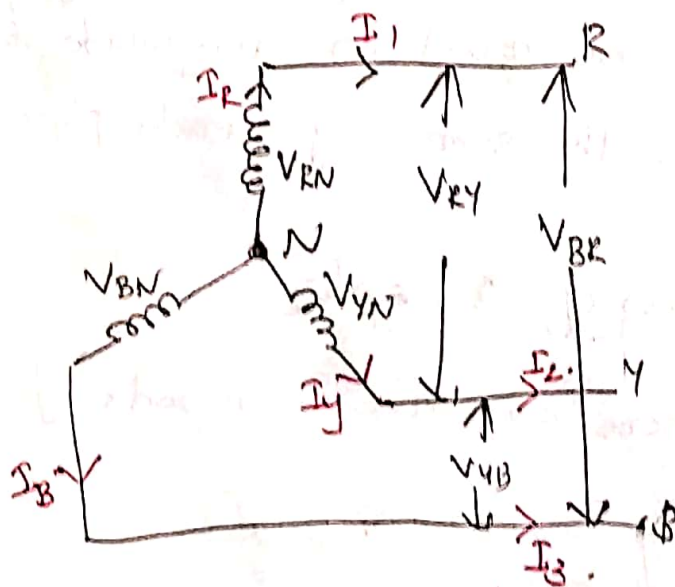
In this method, similar ends of three phases are joined together to form a common junction N called Star or Neutral point. The three line conductors are run from the three ends and are designed as R, Y, B.

The voltage between any line and neutral point is voltage across each winding is called phase voltage while voltage between any two lines is called line currents.

When the load across the system is balanced (i.e. when the emfs, currents and phase angles are equal for each phase), neutral wire carries no current.

$$\therefore I_N = 0 = I_R + I_Y + I_B$$

(I_R, I_Y, I_B are at 120° phase difference with each other).

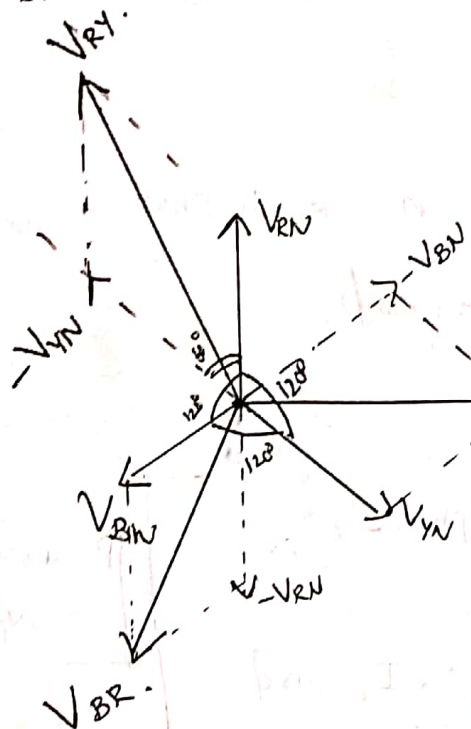


Relation between line and phase voltages

$$V_{RY} = V_{RN} - V_{YN}$$

||y $V_{YB} = V_{YN} - V_{BN}$
 $V_{BR} = V_{BN} - V_{RN}$

R --- N --- Y.



$$V_{RY} = \sqrt{V_{RN}^2 + V_{YN}^2 + 2 V_{RN} V_{YN} \cos 60^\circ}$$

$$V_L = \sqrt{V_{ph}^2 + V_{ph}^2 + 2 V_{ph}^2 \cos 60^\circ}$$

$$V_{YB} = \sqrt{2 V_{ph}^2 + 2 V_{ph}^2 \cos 60^\circ}$$

$$= \sqrt{2 V_{ph}^2 + 2 V_{ph}^2 \times \frac{1}{2}}$$

$$= \sqrt{2 V_{ph}^2 + V_{ph}^2}$$

$$= \sqrt{3 V_{ph}^2}$$

$$= \sqrt{3} V_{ph}$$

$$\therefore \boxed{V_L = \sqrt{3} V_{ph}}$$

Hence in a balanced 3 ϕ star connection.

$$\text{line voltage} = \sqrt{3} \times \text{phase voltage}$$

All line voltages are equal in magnitude but displaced by 120° from one another.

Relation between line and phase currents

In star connection, each line conductor is connected in series to a separate phase.

∴ Current in a line conductor is same as that in phase to which the line conductor is connected.

$$I_L = I_{ph}.$$

Power

Total power = 3 × power in each phase.
 $= 3 \times V_{ph} I_{ph} \cos \phi.$

For star connection

$$V_{ph} = \frac{V_L}{\sqrt{3}} ; I_{ph} = I_L.$$

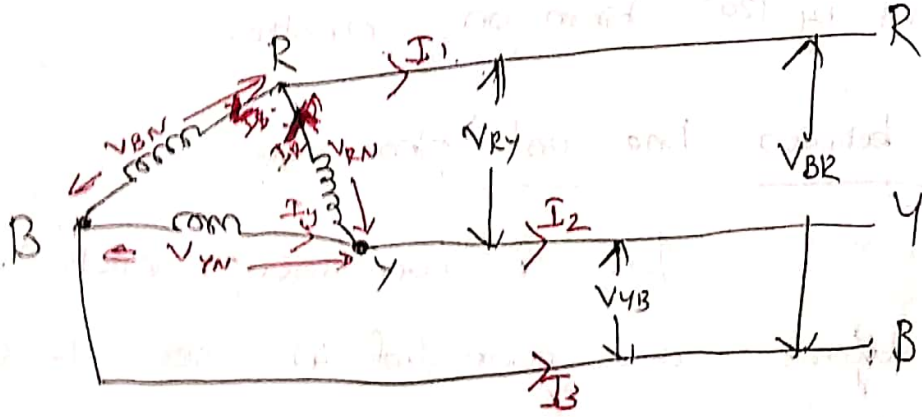
$$\text{Total power} = 3 \times \frac{V_L}{\sqrt{3}} \times I_L \cos \phi$$

Delta Connection $= \sqrt{3} V_L I_L \cos \phi.$

$$P = \sqrt{3} V_L I_L \cos \phi$$

$$= 3 V_{ph} I_{ph} \cos \phi$$

In delta connection, the dissimilar ends of three phase windings are joined together. i.e. finishing end of one phase is connected to starting end of other phase winding & so on.



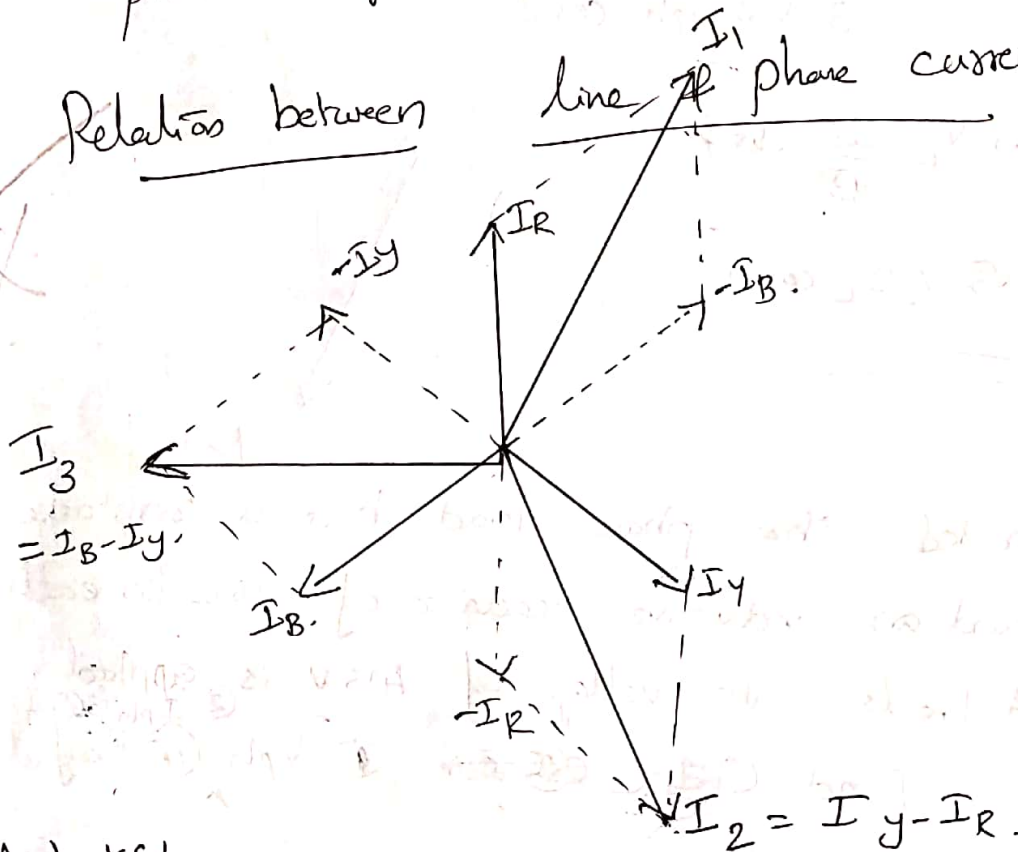
Here V_{RY} , V_{YB} , V_{BR} represents line voltage.

Relation between line & phase voltage

Since only one phase winding is connected across two lines, the line voltage (V_L) is equal to phase voltage (V_{ph})

$$V_L = V_{ph}$$

Relation between line & phase current



Apply KCL

$$I_1 + I_B = I_R$$

$$I_1 = I_R - I_B$$

|||y

$$I_2 = I_Y - I_R \quad \text{f} \quad I_3 = I_B - I_Y$$

$$I_1 = I_R - I_B$$

$$= \sqrt{I_R^2 + I_B^2 + 2I_R I_B \cos 60}$$

$$= \sqrt{I_{ph}^2 + I_{ph}^2 + 2I_{ph}^2 \times \frac{1}{2}}$$

$$= \sqrt{3I_{ph}^2}$$

$$I_1 = \sqrt{3} I_{ph}$$

In Δ connection, line current is equal to $\sqrt{3}$ times phase current.

Power in Delta connected system

$$\begin{aligned} \text{Total power, } P &= 3 \times \text{power in each phase} \\ &= 3 V_{ph} I_{ph} \cos \phi. \end{aligned}$$

$$P = 3 V_L \frac{I_L}{\sqrt{3}} \cos \phi$$

$$P = \sqrt{3} V_L I_L \cos \phi.$$

1. A star connected three phase load has a resistance of 12Ω and an inductive reactance of 16Ω in each branch. A line to line voltage of $415V$ is applied across it. Find
- (1) ~~I_L~~
 - (2) I_{ph}
 - (3) I_L
 - (4) V_{ph}
 - (5) Total power.

Soln,

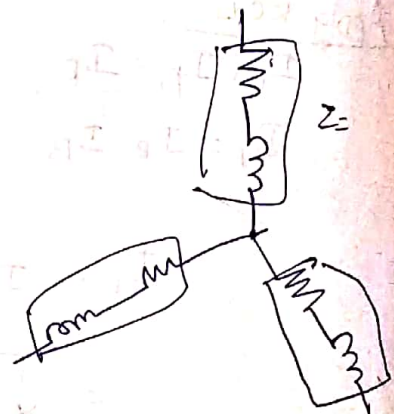
$$R = 12\Omega, X_L = 16\Omega$$

$$V_L = 415V.$$

$$(1) V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{415}{\sqrt{3}} = 239.6V$$

$$(2) \text{ phase current, } I_{ph} = \frac{V_{ph}}{Z} = \frac{239.6}{\sqrt{12^2 + 16^2}} = 11.98A$$

$$(3) \text{ Line current, } I_L = I_{ph} = 11.98A.$$



(14) Power factor, $\cos \phi = \frac{R}{Z} = \frac{12}{\sqrt{12^2 + 16^2}} = 0.6$

Total power, $P = 3 V_{ph} I_{ph} \cos \phi$
 $= 3 (239.6) (11.98) 0.6$
 $= \underline{\underline{5166.73 W}}$

2. Three impedance of $(4 + 3j) \Omega$ are connected in delta across three phase 200V supply. Find line current, power factor and active power -

Soln $Z_{ph} = \sqrt{4^2 + 3^2} = 5 \Omega$

For delta, $V_L = V_{ph} = 200 V$

$I_{ph} = \frac{V_{ph}}{Z} = \frac{200}{5} = 40 A$

$I_L = \sqrt{3} I_{ph} = \sqrt{3} \times 40 = 69.3 A$

Power factor $\cos \phi = \frac{R}{Z} = \frac{4}{5} = 0.8$

Active power, $P = \sqrt{3} V_L I_L \cos \phi$
 $= \sqrt{3} \times 200 \times 69.3 \times 0.8$
 $= 19204 W //$
 $= \underline{\underline{19.2 kW}}$